For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory.

ALTIVA-D

Extended Release Tablets

(Fexofenadine HCl & Pseudoephedrine Sulphate)

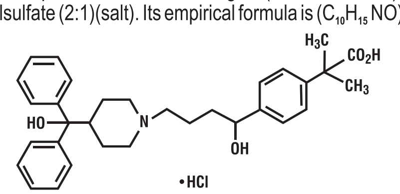
COMPOSITION
Each film coated tablet contains

Fexofenadine Hydrochloride	120 mg
Pseudoephedrine Sulphate USP	180 mg

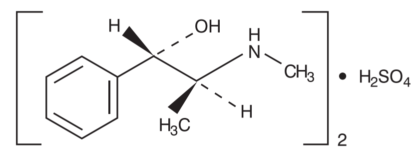
(As Extended Release)

PRODUCT DESCRIPTION
Round, biconvex, film coated, bilayer tablet with one yellow coloured layer and other white to off-white coloured layer.

DESCRIPTION
Fexofenadine hydrochloride is a histamine H1-receptor antagonist and is chemically designated as (±)-4-[1-hydroxy-4-[4-(hydroxydiphenylmethyl) -1-piperidinyl]- butyl]-a, a-dimethylbenzeneacetic acid hydrochloride. Its empirical formula is $C_{32}H_{39}NO_4$ .HCl and its molecular weight is 538.13.
Pseudoephedrine sulphate is an adrenergic (vasoconstrictor) agent and is chemically designated as a -[1-(methylamino)ethyl]-[S-(R*,R*)]-benzenemethanolsulfate (2:1)(salt). Its empirical formula is $(C_{10}H_{15}NO)_2$, H_2SO_4 and its molecular weight is 428.54.



FEXOFENADINE HYDROCHLORIDE



PSEUDOEPHEDRINE SULPHATE

STRUCTURAL FORMULAE

PHARMACOLOGY
Mechanism of Action
Fexofenadine, a metabolite of terfenadine, is an anti-histamine with selective peripheral H1 - receptor antagonist activity. It lacks anti-cholinergic or alpha1 - adrenergic receptor blocking activities. Pseudoephedrine sulphate exerts its effects by releasing adrenergic mediators from postganglionic nerve terminals. It brings about its nasal decongestant action by activation of alpha-adrenergic receptors in venous capacitance vessels in nasal tissues. Pseudoephedrine sulphate has little or no pressor effects in normotensive adults.
Altiva-D extended release tablets should be administered when both the antihistaminic properties of Fexofenadine and the nasal decongestant activity of Pseudoephedrine are required.

Pharmacokinetics
The pharmacokinetics of fexofenadine hydrochloride and pseudoephedrine sulphate when administered separately have been well characterised.
Fexofenadine is rapidly absorbed after oral administration with mean peak plasma concentration occurring approximately 1-3 hours after a single dose. Administration with food reduces the peak plasma concentration (Cmax) and area under the curve (AUC0-a) of fexofenadine and delays the time to achieve peak plasma concentration (Tmax). Fexofenadine is 60% to 70% bound to plasma proteins, primarily to albumin and a-1-acid glycoprotein. Fexofenadine does not cross the blood brain barrier. The metabolism of fexofenadine is minimal with approximately 5% of the dose being metabolised to a ketone-acid metabolite. Approximately 11% of an oral dose of fexofenadine is excreted in the urine, primarily as unchanged drug and 80% is excreted in faeces. The mean elimination half-life of fexofenadine is about 14.4 hours. In the histamine-wheal test in healthy human volunteers, the administration of a single dose of **Altiva-D** extended release tablet (containing Fexofenadine 120mg and 180mg Pseudoephedrine sulphate) showed a maximum mean percentage decrease in the wheal area of 84.31± 3.53 at 4.33±0.33 hours and maximum mean percentage decrease in flare area of 89.74± 1.63 at 4.83± 0.30 hours.
Pseudoephedrine is well absorbed after oral administration. Presence of food in the stomach does not alter the absorption of pseudoephedrine. Following the administration of a single dose of **Altiva-D** extended release tablet (containing Fexofenadine 120mg and 180mg Pseudoephedrine sulphate) in healthy human volunteers, peak plasma concentration (Cmax) of 560± 44.2 ng/ml was achieved at 6.92± 0.57 hours (Tmax) for the pseudoephedrine component. The AUC (0-a) value was 5528.19± 368.70 ng/ml x hour. Plasma protein binding is negligible for pseudoephedrine. The plasma elimination half-life is about 4-6 hours. Pseudoephedrine is excreted unchanged in the urine.
The bioavailability of fexofenadine and pseudoephedrine from **Altiva-D** extended release tablets is similar to that achieved following separate administration of each component. Co-administration of fexofenadine and pseudoephedrine does not significantly affect the bioavailability of either component.

INDICATIONS
ALTIVA-D is indicated for the relief of symptoms associated with seasonal allergic rhinitis including rhinorrhea, sneezing, itchy watery red eyes, itchy nose, palate and/or throat, and nasal congestion. It should be administered when both the antihistaminic properties of fexofenadine and nasal decongestant activity of pseudoephedrine are desired.

DOSAGE AND ADMINISTRATION
The recommended dose of **Altiva-D** is one tablet for adults and children 12 years of age and older. It is recommended that administration of **Altiva-D** with food should be avoided.
Since lower doses are recommended for patients with renal failure, it would not be appropriate to recommend **ALTIVA-D**.

PRECAUTIONS

i) **General**
Due to its pseudoephedrine component, **Altiva-D** should be used with caution in patients with hypertension, diabetes mellitus, ischaemic heart disease, increased intraocular pressure, hyperthyroidism, renal impairment or prostatic hypertrophy.

ii) **Warning**
Central nervous system stimulation with convulsion or cardiovascular collapse with accompanying hypotension may be produced by sympathomimetic amines.
The elderly are more likely to have adverse reactions to sympathomimetic amines.

iii) **Contraindications**

Altiva-D is contraindicated in patients with a known hypersensitivity to any of its ingredients.

This product, due to its pseudoephedrine content is contraindicated in patients with narrow angle glaucoma or urinary retention, and in patients receiving monoamine oxidase (MAO) inhibitor therapy or within 14 days of stopping such treatment. It is also contraindicated in patients with severe hypertension, severe coronary artery disease and in those who have shown hypersensitivity or idiosyncrasy to its components, to adrenergic agents, or to other drugs of similar chemical structures. Manifestations of patient idiosyncrasy to adrenergic agents include; insomnia, dizziness, weakness, tremor or arrhythmias.

iv) **Pregnancy**

Adequate and well controlled studies have not been performed in humans with fexofenadine and pseudoephedrine combination. **Altiva-D** should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

v) **Lactation**

It is not known if fexofenadine is excreted in human milk. Because many drugs are excreted in human milk, caution should be used when fexofenadine hydrochloride is administered to a nursing woman. Pseudoephedrine administered alone distributes into the breast milk of lactating human females. Caution should be exercised when **Altiva-D** is administered to nursing women.

vi) **Paediatrics**

Safety of **Altiva-D** in paediatric patients under the age of 12 years has not been established.

vii) **Geriatrics**

Since elderly patients are more likely to have decreased renal function, necessary care should be taken and it may be useful to monitor renal function.

viii) **Carcinogenicity / Mutagenicity/ Impairment of fertility**

There are no animal or *in vitro* studies on the combination of fexofenadine and pseudoephedrine to evaluate carcinogenesis, mutagenesis or impairment of fertility. Studies in mice and rats with terfenadine have not shown any evidence of tumorigenicity. Micronucleus test assays of fexofenadine have not shown any evidence of mutagenicity.

ix) **Drug Interactions**

Fexofenadine and pseudoephedrine do not influence the pharmacokinetics of each other when administered concomitantly.

Erythromycin and ketoconazole increase the concentrations of co-administered fexofenadine, however, no increase in adverse events or QTc interval have been reported when subjects were administered fexofenadine alone or in combination with erythromycin or ketoconazole. The changes in plasma levels were within the range of plasma levels achieved in adequate and well-controlled clinical trials. The mechanism of these interactions has been evaluated in *in vitro*, *in situ* and *in vivo* animal models. These studies indicate that ketoconazole or erythromycin co-administration enhances fexofenadine gastrointestinal absorption. *In vivo* animal studies also suggest that in addition to enhancing absorption, ketoconazole decreases fexofenadine gastrointestinal secretion, while erythromycin may also decrease biliary excretion. Fexofenadine has no effect on the pharmacokinetics of erythromycin and ketoconazole.

Altiva-D tablets (pseudoephedrine component) are contraindicated in patients taking monoamine oxidase (MAO) inhibitors and for 14 days after stopping use of an MAO inhibitor.

Concomitant use with antihypertensive drugs which interfere with sympathetic activity (eg. methyldopa, mecamylamine and reserpine) may reduce their antihypertensive effects. Increased ectopic activity can occur when pseudoephedrine is used concomitantly with digitalis.

Care should be taken in the administration of **Altiva-D** concomitantly with other sympathomimetic amines because the combined effects on the cardiovascular system may be harmful to the patient.

X) **Adverse Effects**

Adverse events in patients with seasonal allergic rhinitis receiving fexofenadine and pseudoephedrine combination have been reported to be similar to those reported either in patients receiving fexofenadine or patients receiving pseudoephedrine alone. Side effects reported in clinical trials with fexofenadine are mild and infrequent. The side effects reported include headache, drowsiness, nausea, dizziness and fatigue.

Pseudoephedrine may cause symptoms of CNS stimulation: sleep disturbances and, rarely, hallucinations have been reported. Skin rashes with or without irritation have occasionally occurred. Urinary retention has ben reported occasionally in men receiving pseudoephedrine; prostatic enlargement could have been an important predisposing factor.

OVERDOSAGE

Most reports of fexofenadine hydrochloride overdose contain limited information. However, dizziness, drowsiness and dry mouth have been reported. In the event of overdose, standard measures should be adopted to remove any unabsorbed drug. Symptomatic and supportive treatment is recommended.

SHELF LIFE : 24 months.

STORAGE

Store below 30°C, protected from light and moisture.

SUPPLY

Strip of 10 tablets. Box of 10X 10's.

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

Date of Revision: January 2021

Manufactured in India by:
Sun Pharmaceutical Ind. Ltd.
Industrial Area - 3
Dewas - 455 001

Product Registration Holder:
RANBAXY (MALAYSIA) SDN. BHD.
(A Sun Pharma Company)
Lot 23, Bakar Arang Industrial Estate
08000 Sungai Petani, Kedah, MALAYSIA